

Data Flow Diagram

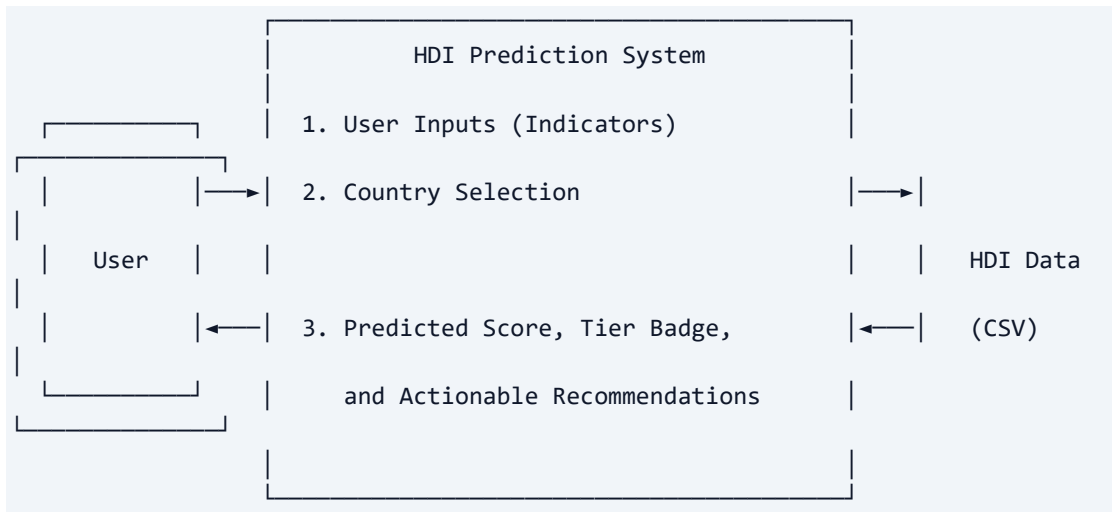
DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	2 MARKS

1. Overview

The Data Flow Diagram (DFD) maps the flow of data within the Human Development Index (HDI) Prediction System, detailing how user inputs are processed, validated, and mapped to outputs.

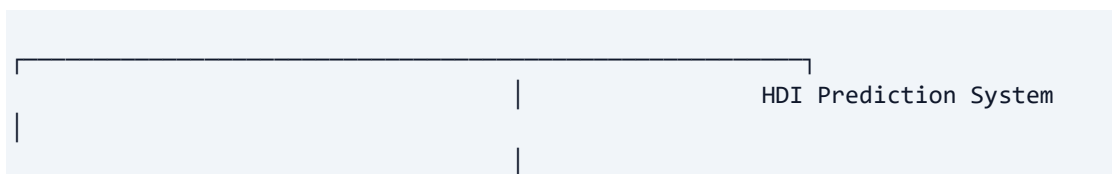
2. Level 0: Context DFD

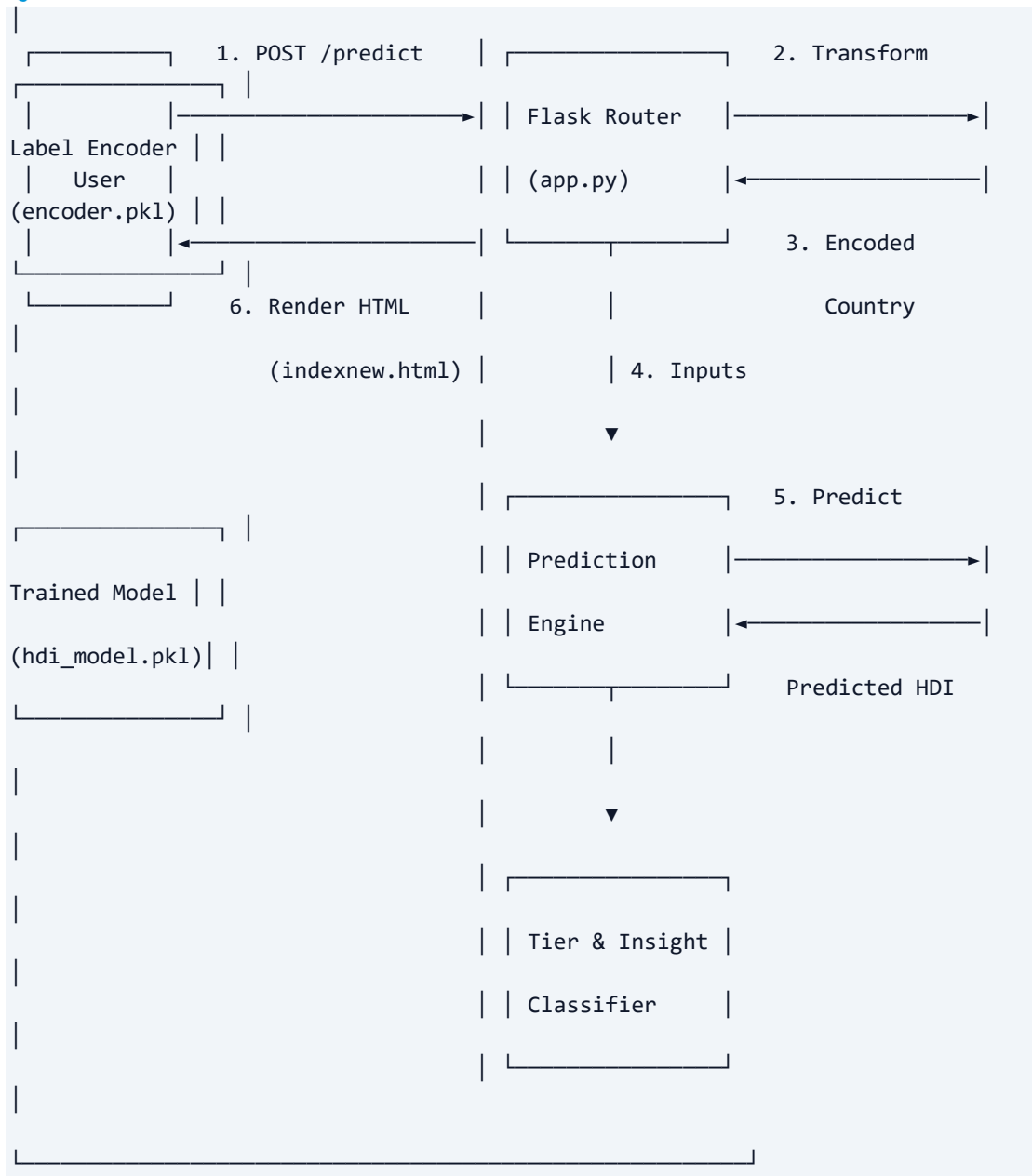
The Level 0 Context Diagram shows the system's boundary, external entities (User, Dataset), and high-level input/output flows.



3. Level 1: System Processes

The Level 1 DFD decomposes the system into core functional processes: data loading/training, route handling, prediction, classification, and result rendering.





4. Level 2: Pipeline Data Flow

4.1 Model Training Pipeline (ml_model.py)

- Data Load:** Load raw data from `data/hdi\_data.csv` using Pandas.
- Feature Extraction:** Select features: `Country`, `Life\_Expectancy`, `Expected\_Years\_Schooling`, `Mean\_Years\_Schooling`, `GNI\_Per\_Capita`.
- Imputation:** Handle missing values using mean imputation.
- Encoding:** Fit and save the `LabelEncoder` (`models/encoder.pkl`) to convert country categories to integers.

5. **Data Split:** Partition dataset into 80% training and 20% test splits.
6. **Fit & Score:** Train the Linear Regression model, evaluate using R^2 and RMSE metrics, and serialize the trained model (`models/hdi\_model.pkl`).

4.2 Prediction Pipeline (app.py)

7. **HTTP Form Read:** Read request inputs: `country`, `life\_expectancy`, `expected\_years\_schooling`, `mean\_years\_schooling`, `gni\_per\_capita`.
8. **Type Cast:** Convert inputs to numeric values.
9. **Encode Input:** Transform the country category name using the loaded `LabelEncoder`.
10. **Execute Model:** Pass features to the loaded Linear Regression model to obtain the raw prediction score.
11. **Post-Process:** Clamp prediction score to a range of `[0.0, 1.0]`.
12. **Classify Tier:** Determine the development tier based on standard thresholds.
13. **Generate Insights:** Fetch insights and recommendations matching the predicted tier.
14. **Render Template:** Return `indexnew.html` with score, badge color, progress gauge position, and recommendation list.

5. Data Dictionary

Feature Name	Type	Description	Boundary Constraints
`country`	Categorical	Target country class category	Category A to E
`life_expectancy`	Float	Average life expectancy at birth	50.0 to 85.0 years
`expected_years_schooling`	Float	Expected school duration for kids	5.0 to 20.0 years
`mean_years_schooling`	Float	Mean school duration for adults	3.0 to 15.0 years
`gni_per_capita`	Float	Gross National Income per Capita	\$500.00 to \$75,000.00